

MSDM 5001

An Introduction to Computational and Modelling Tools

Lecture 11: Introduction to Quantum Computing – Part II

Topics to discuss

- Quantum computers vs classical computers
- The circuit model for quantum computing
 - 1 qubit gates
 - Multi-qubit gates
 - The notion of “universal” gates
 - Measuring complexity in quantum computing

Warning: Today will be a bit more math heavy!

What is a quantum computer?



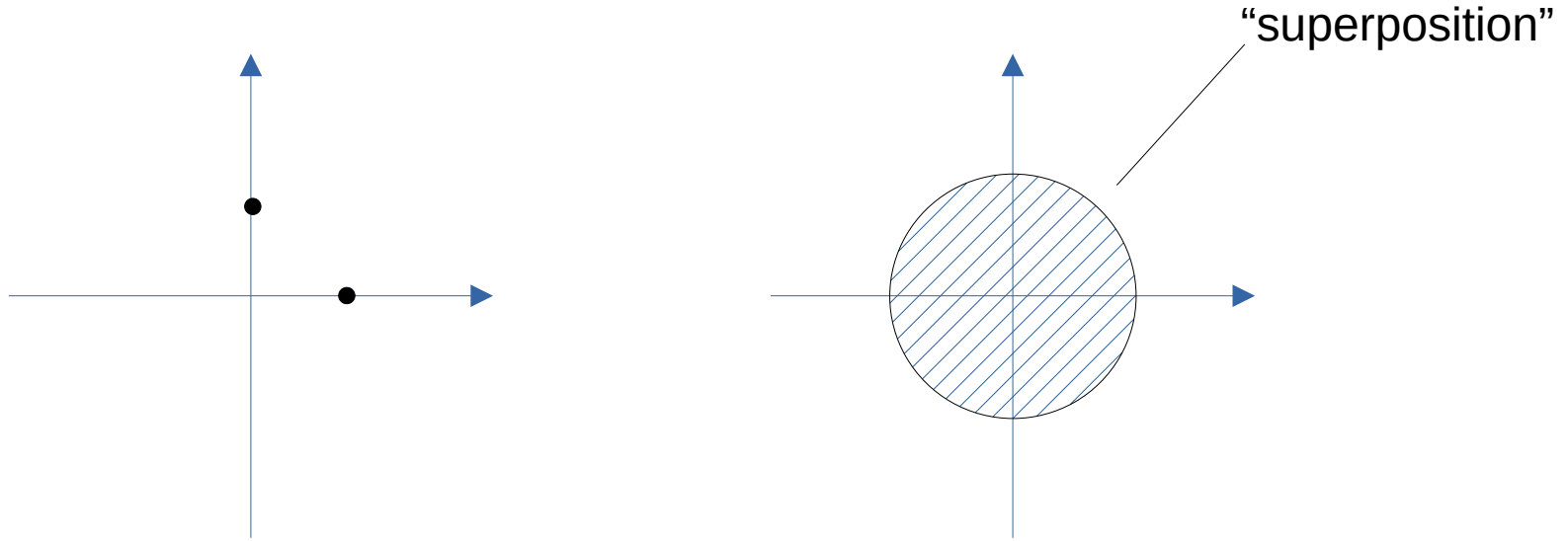
Note: The extra qubits are called “ancillary”. More on them later

Layers of abstractions

	Classical (digital) computers	Quantum computers
Actual data	int, float, string, graph, ...	
Representation	bits	qubits
Physical realization	Currents/voltages in circuits	• Photon polarizations • Energy states of atoms • Superconductors (cooper pairs) • ...

Quantum vs classical computers?

- Why **might** a quantum computer be faster?



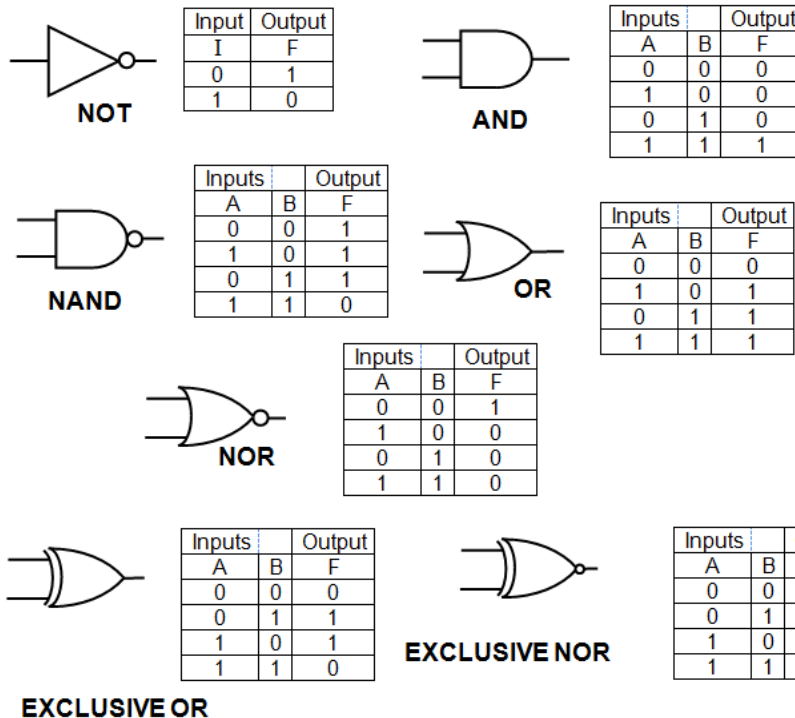
“state space” of one classical bit vs one quantum bit

Quantum vs classical computers?

- Why might a quantum computer be faster?
- Claim:
For a two-qubit system, there are states which cannot be expressed as tensor product of two one-qubit states
 - “Entanglement”
- In other words, quantum mechanics give new computational resources not possible in classical computing!
 - Interference & entanglement
 - But just because you have more “resources” doesn’t mean it’s easy to make use of them.....

The circuit model of quantum computing

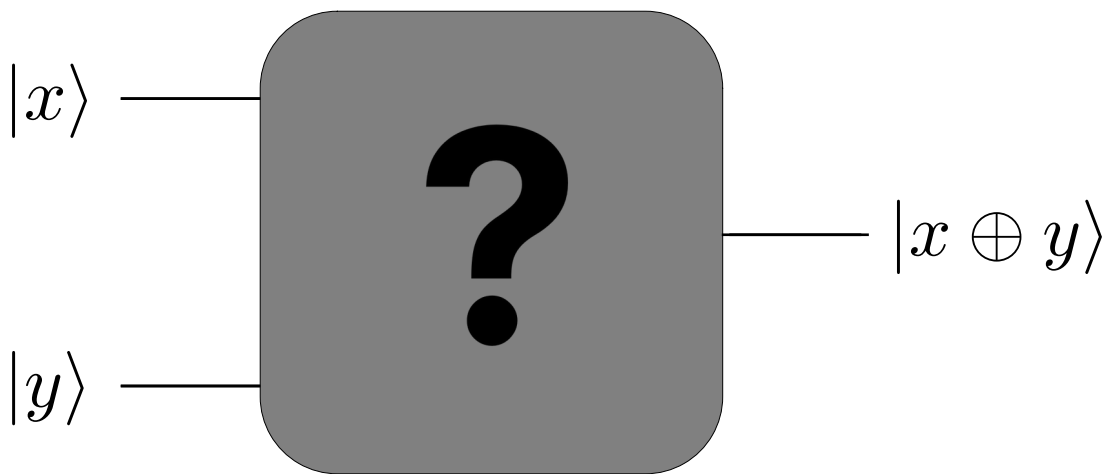
- Classical computers



Gates: basic operations that take in bits and output processed bits

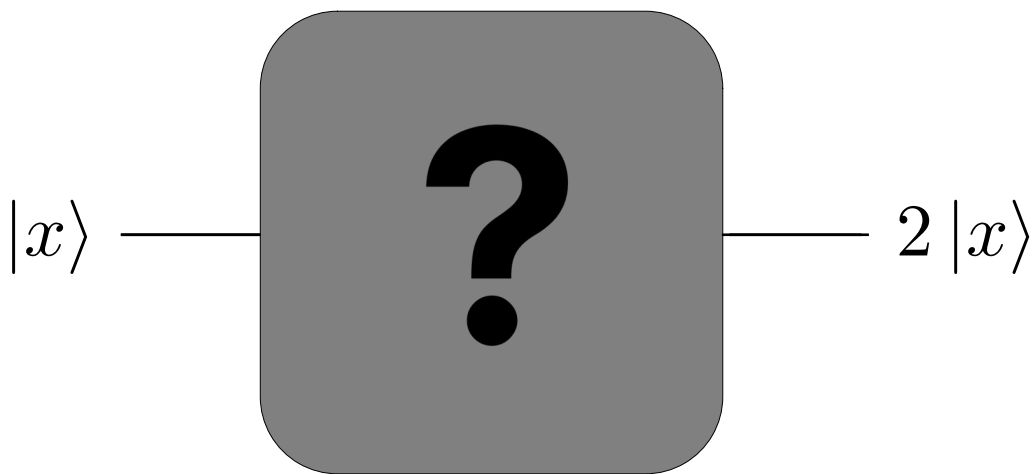
The circuit model of quantum computing

- Is this “quantum gate” possible?



The circuit model of quantum computing

- How about this?



Interlude: The Schrodinger's Equation

- The equation of motion in QM (c.f. Newton's 2nd Law) is

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle$$

where H is hermitian (the “Hamiltonian”)

- Formally, the solution to this differential equation is (assuming time-independent Hamiltonian)

$$|\psi(t)\rangle = \exp(-iHt/\hbar) |\psi(0)\rangle \equiv U(t) |\psi(0)\rangle$$

Interlude: The Schrodinger's Equation

- One can show that $U(t)$ is unitary:

$$U^\dagger U = \mathbb{1}$$

- In fact, by requiring time-evolution to be norm-preserving and information-preserving, one can derive the Schrodinger's equation! *

* Also need to demand a group structure

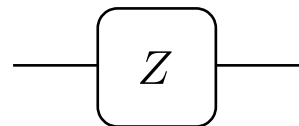
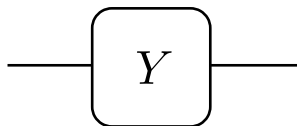
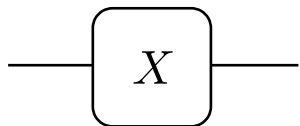
The circuit model of quantum computing

- Conclusion
 - Quantum gates must be unitary!
 - In other words, quantum computing must be “reversible”
 - This explains those “ancillary” qubits we saw before

One-qubit gates

- Some examples (in computational basis)

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



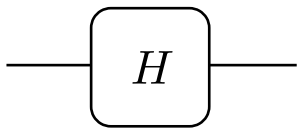
- Note: They are basically the Pauli matrices

One-qubit gates

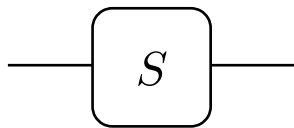
- Some more examples (in computational basis)

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

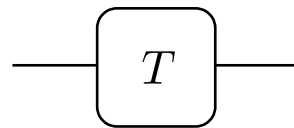
“Hadamard”



“Phase”



“ $\pi/8$ ”



Interlude: Bloch sphere

- It is possible to visualize the actions of one-qubit gates
- Any given one-qubit state takes the form

$$|q\rangle = a |0\rangle + be^{i\phi} |1\rangle \quad a, b \in \mathbb{R}, \phi \in [0, 2\pi)$$

- Why this form?

- Normalization requires

$$a^2 + b^2 = 1$$

- So we can choose to parametrize as

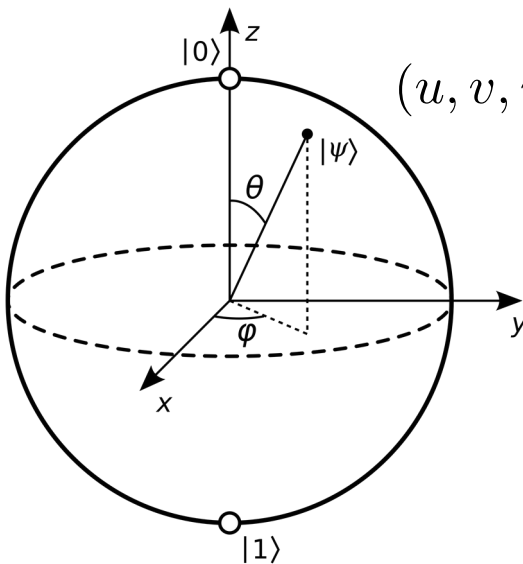
$$a = \cos\left(\frac{\theta}{2}\right), \quad b = \sin\left(\frac{\theta}{2}\right) \quad \theta \in [0, \pi]$$

Interlude: Bloch sphere

- In conclusion, we have a one-to-one mapping

$$|q\rangle \leftrightarrow (\theta, \phi)$$

- This should remind you of spherical coordinates...



$$(u, v, w) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

“Bloch vector”

Exercise 1: Bloch sphere

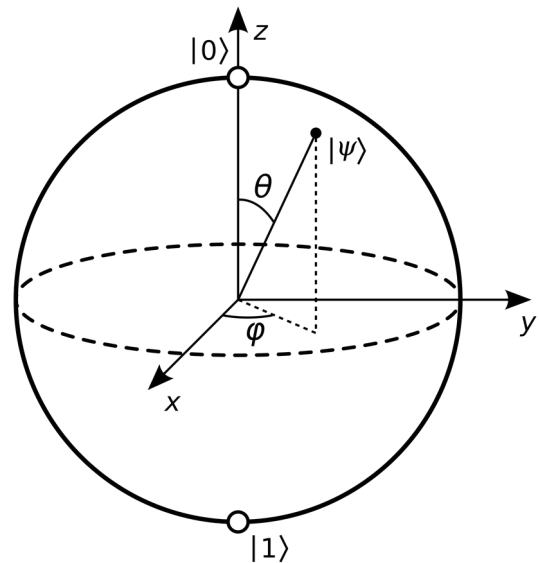
- How does the X gate look under this representation?

$$X : \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \mapsto \begin{pmatrix} \beta \\ \alpha \end{pmatrix}$$

$$\theta \mapsto ???$$

$$\phi \mapsto ???$$

Geometrically, this is _____



One-qubit gates

- Even more examples

$$R_x(\theta) = e^{-\theta X/2}, \quad R_y(\theta) = e^{-\theta Y/2}, \quad R_z(\theta) = e^{-\theta Z/2}$$

- What does it mean to exponentiate a matrix?? 🤔

$$e^A := \sum_{n=0}^{\infty} \frac{A^n}{n!}$$

One-qubit gates

- Even more examples

$$R_x(\theta) = e^{-\theta X/2}, \quad R_y(\theta) = e^{-\theta Y/2}, \quad R_z(\theta) = e^{-\theta Z/2}$$

- What do these gates do in terms of the Bloch sphere?

Universal one-qubit gates

- Claim: Any 2 by 2 unitary matrix can be written as

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta)$$

for some $\alpha, \beta, \gamma, \delta \in \mathbb{R}$

- In other words, $\{R_y, R_z\}$ are “universal” in the sense that any other one-qubit gates can be built from them
- In group theory language, they are called the “generators” of $U(2)$

Multi-qubit gates

- As we saw, they must have same number of input and output gates
- Example: CNOT

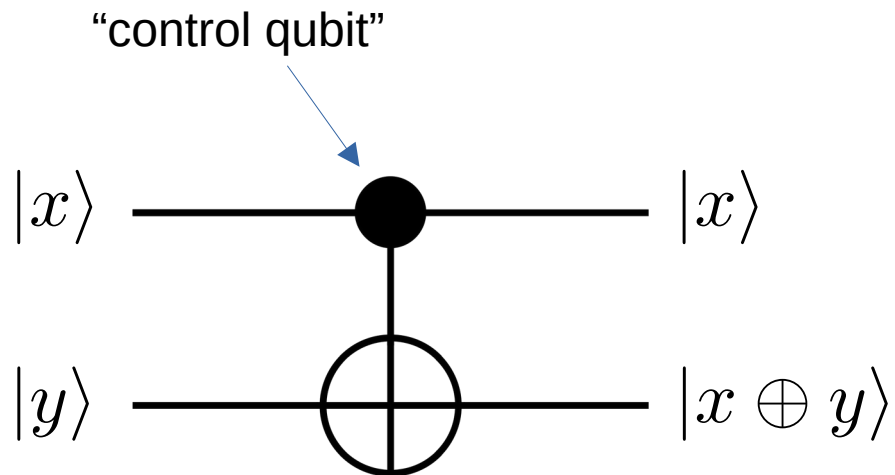
“Flip the second qubit if the first qubit reads 1, else keep it the same”

$$|00\rangle \mapsto |00\rangle$$

$$|01\rangle \mapsto |01\rangle$$

$$|10\rangle \mapsto |11\rangle$$

$$|11\rangle \mapsto |10\rangle$$



Check that CNOT is unitary

Multi-qubit gates

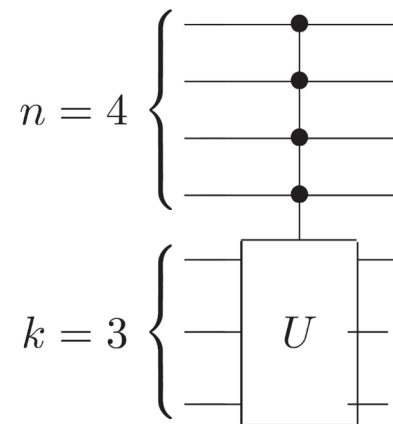
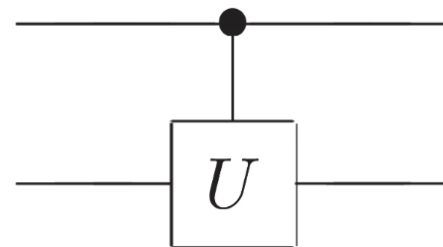
- In general, one can define a “control-U” gate by

$$|0q\rangle \mapsto |0q\rangle$$

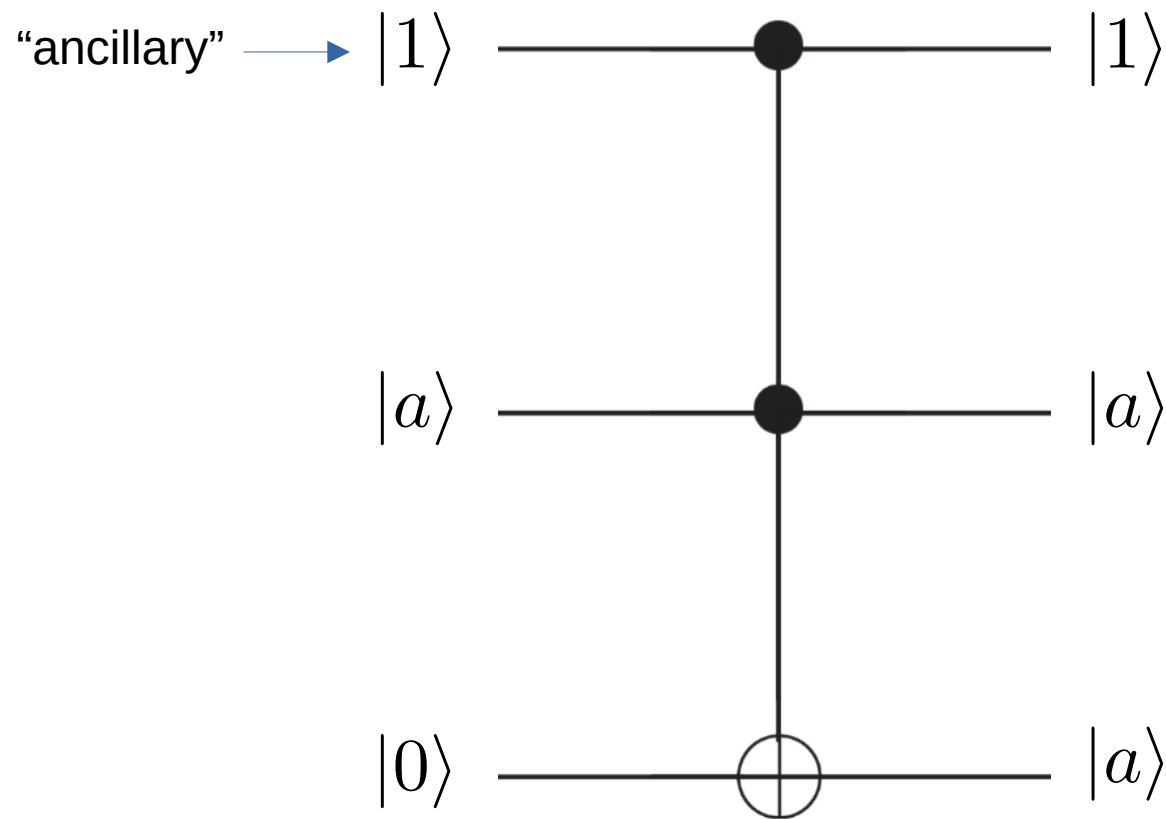
$$|1q\rangle \mapsto |1\rangle \otimes (U |q\rangle)$$

- Can even have more control qubits:

$$\begin{aligned} C^n(U) |c_1 \dots c_n q_1 \dots q_k\rangle \\ = |c_1 \dots c_n\rangle \otimes U^{c_1 \dots c_n} |q_1 \dots q_k\rangle \end{aligned}$$

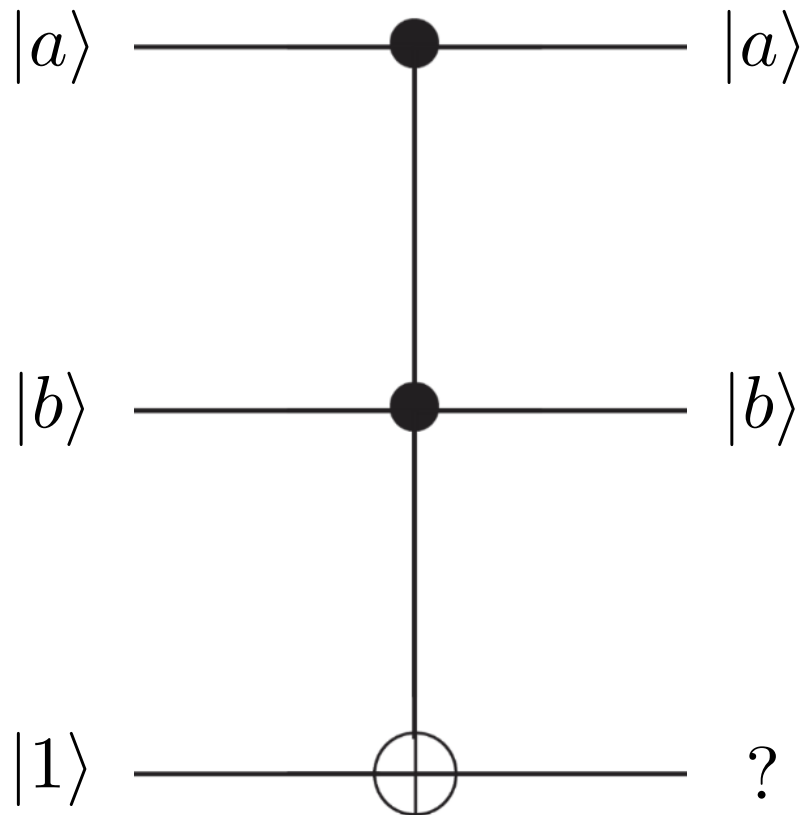


Example: Toffoli gate



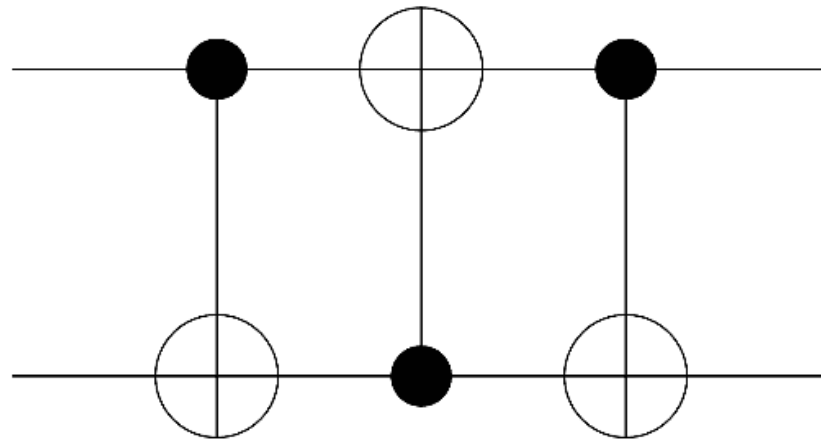
We have copied a qubit!

Example: Toffoli gate



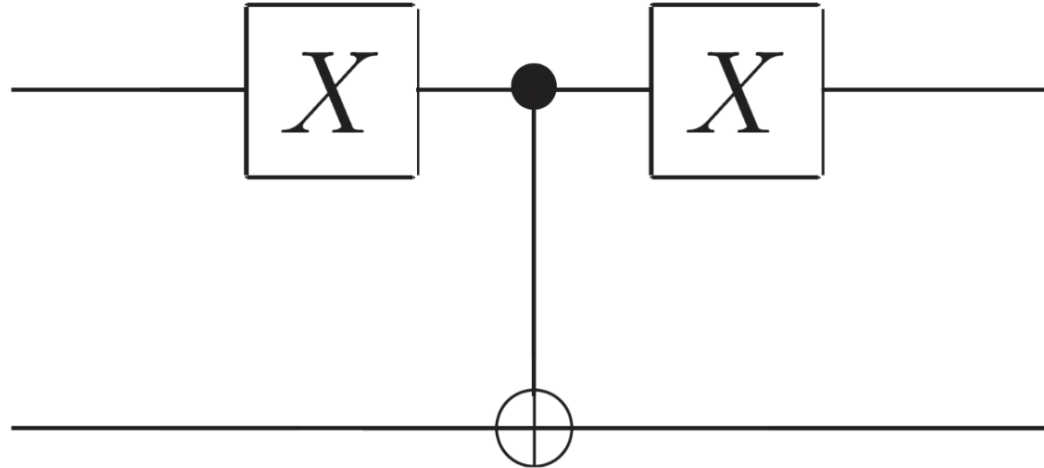
Exercise 2

- Analyze the effects of this quantum circuit:



Exercise 2

- What about this?

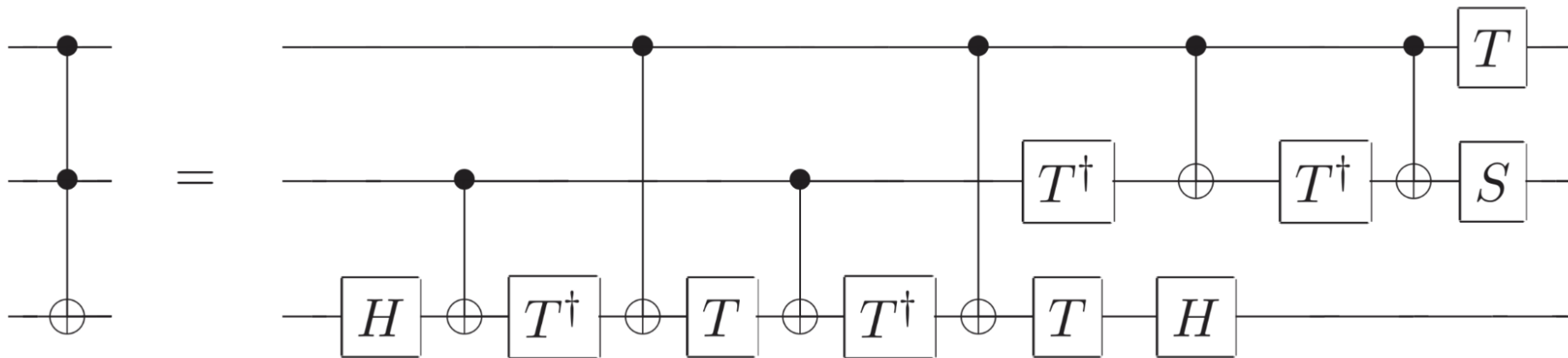


General universal gates

- Claim: Any unitary operators can be approximated to arbitrary precision by

$$\{H, S, T, \text{CNOT}\}$$

- Example:



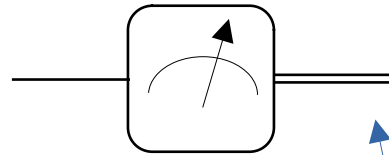
General universal gates

- Comments
 - There are infinite families of sets of universal gates. Exactly which one is “better” depends on the actual physical realization
 - Universality $\neq$ Efficiency
Most unitary gates requires an exponential number of universal gates to approximate!
 - Efficiency in quantum computing: given a problem of size n , how does the circuit used to solve it scale with n ?
 - Therefore, one difficulty of quantum computing is constraining ourselves to use only unitary operators that can be efficiently implemented (hardware-limited)

Measurements

- The final element we need is the act of measurement
 - Ultimately we have to read the output of a quantum circuit to do anything useful!

Measurement in the **computational basis**



Notice the double line!
Why don't I just use a single line? 🤔

Measurements

- Some comments:
 - Generally speaking, measurements destroy quantum information and is an irreversible (i.e. non-unitary) process
 - But if everything is described by the rules of QM and in particular Schrodinger's equation, how is this possible?????
 - This is (part of) the so-called measurement problem. Answering it is beyond the scope of this course. Look up the term “decoherence” if you are interested.

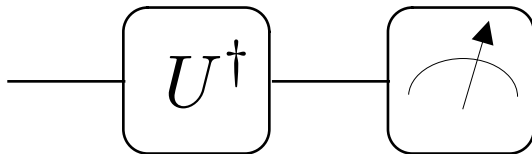
Measurements

- What if we want to measure in another basis?

$$|v_0\rangle = U |0\rangle$$

$$|v_1\rangle = U |1\rangle$$

- Claim: The following circuit does what we want



References

- Quantum Computation and Quantum Information, Nielsen and Chuang, Chapter 4